Operational Research: Tutorial 2 Answers

Linear Programming 1

The more complicated questions on the tutorial are marked by:

* Medium

** Hard

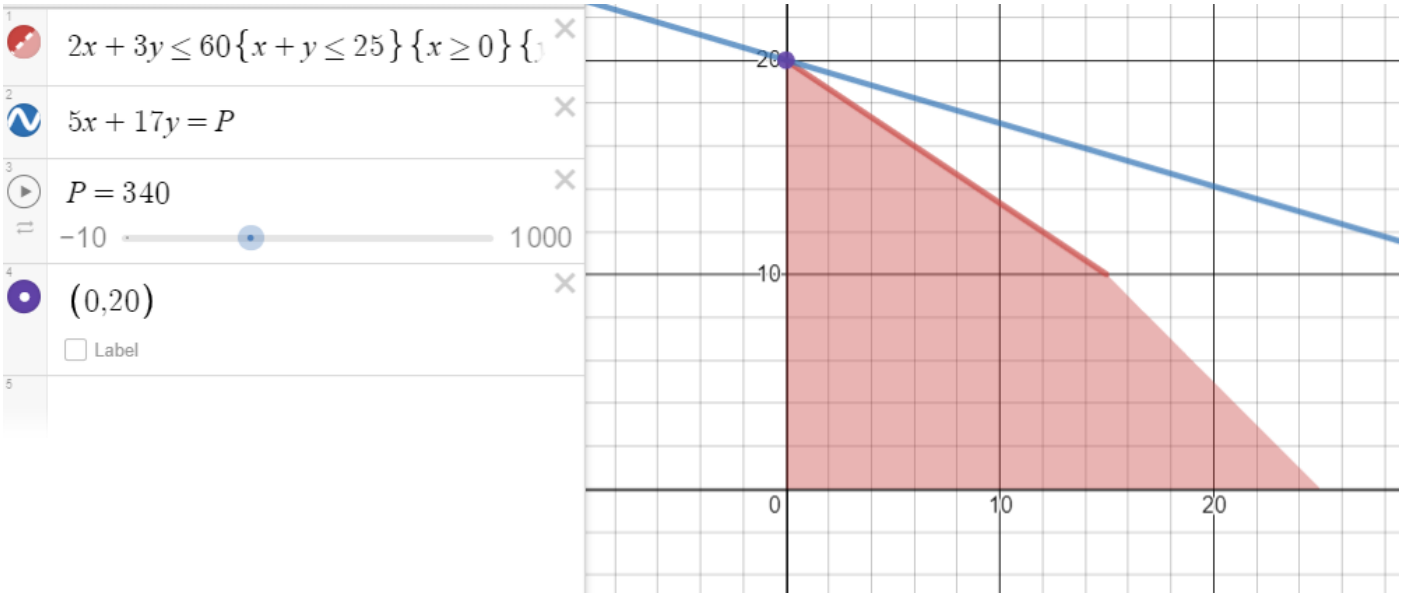
Exercises

1) Graphically solve the following linear programming (LP) problem

maximise $5x + 17y$

subject to:

$$2x + 3y \leq 60$$
$$x + y \leq 25$$
$$x, y \geq 0$$



2) Express the LP problem in matrix form.

maximise $\begin{pmatrix} 5 & 17 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

subject to:

$$\begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \leq \begin{pmatrix} 60 \\ 25 \end{pmatrix}$$

and:

$$\begin{pmatrix} x \\ y \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

3) Implement the problem in Python and use the `linprog` method to confirm your solution.
Please see the notebook "Q3 LP with linprog".

Sensitivity: Internal

Note that to maximise a function f you need to minimise $-f$

```
Optimal solution found.


x =

     0
    20


fval =

  -340
```

4) You should get the same answer as 3): see the workbook "Q4 LP with Excel Solver".

5) A firm manufactures 2 products, X and Y. *

a)

- **Each one is processed by a single machine, machine A, which has a capacity of 100 hours per week.**
- **It takes 5 hours to make product X and 2 hours to make product Y.**
- **The profit made (per unit) for X and Y is £12 and £5 respectively.**

How many X's and Y's should be made per week so that the largest profit is made?

maximise $12x + 5y$

subject to:

$$5x + 2y \leq 100$$
$$x, y \geq 0$$

Please see the notebook "Q5a Manufacturing - products X and Y"; you should get the following output for the answer.

```
x =

     0
    50


fval =

  -250
```

Investigate the following

b) The firm adds a third product Z which takes 3 hours to manufacture and has a profit of £8
The firm only has enough material a week to make 20 of product Z

maximise $12x + 5y + 8z$

subject to:

$$5x + 2y + 3z \leq 100$$

$$z \leq 20$$
$$x, y, z \geq 0$$

Please see the notebook "Q5b Manufacturing - add product Z"; you should get the following output for the answer.

```
x =

         0
   20.0000
   20.0000


fval =

  -260
```

c) In addition to product Z, the firm reduces the time to produce product X to 4.5 hours
What do you notice about the solution, why might this be a non-optimal solution?

Please see the notebook "Q5c Manufacturing - faster X"; you should get the following output for the answer.

```
x =

   22.2222
         0
         0


fval =

 -266.6667
```

If you can only produce whole numbers, then $x = 22$, results in 99 machine hours, no time to make anything else and a profit of £264.

However, $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 21 \\ 1 \\ 1 \end{pmatrix}$, results in a machine time of 99.5 hours and a profit of £265. Whilst this might seem negligible, scale these numbers to the thousands or millions!!

The best whole-number plan earns £266, for example x = 20, y = 2, z = 2, which uses exactly 100 machine hours. Also notice that X and Z both earn £8/3 per machine hour, so the LP has other optimal solutions with the same profit of £266.67 (for example z = 20, x = 8.89); a solver may report any of them. Week 3 shows how to ask for whole numbers directly (integer programming).

6) N&F (Management Consultants) have been employed to help the County dieticians develop attractive, nutritious, and economical diets for the county meals service. They have been employed in response to the many complaints about the palatability of the meals provided by this service in recent times. **

The meals provided must satisfy certain basic restrictions on vitamin and fat content. Table 1 gives the target quantities for a sample of 4 vitamins and one fat type.

The quantities of each nutrient contained in one unit of each of 10 foods and the cost and attractiveness of a unit of each food are given in Table 2.

You are required to:

a. **Obtain the optimal solution given that the aim is to minimise the cost of the diet.**
b. **Obtain the optimal solution given that the aim is to maximise the attractiveness of the diet.**

Sensitivity: Internal

c. Investigate the effects on these solutions if there are maximum and minimum restrictions on both fat and vitamin content where maximum content = minimum content + 20%.

	Minimum	Maximum
Vitamin A	100	-
Vitamin B	120	-
Vitamin C	200	-
Vitamin D	80	-
Fat	-	300

Table 1

Food	Vitamin A	Vitamin B	Vitamin C	Vitamin D	Fat	Cost	Attractiveness
1	18	4	-	6	-	10	6
2	8	6	22	14	-	6	1
3	22	10	-	-	8	8	5
4	6	12	-	6	-	16	5
5	-	14	-	10	-	8	6
6	-	-	20	8	6	3	4
7	8	-	-	6	20	6	10
8	4	-	8	-	40	8	10
9	-	6	10	12	-	12	6
10	12	20	-	4	-	12	7

Table 2

Sensitivity: Internal

a)

Variable x_i – Food item i

Constants c_i – cost of item i

$(Va)_i$ - Vitamin A content for item i

$(Vb)_i$ - Vitamin B content for item i

$(Vc)_i$ - Vitamin C content for item i

$(Vd)_i$ - Vitamin D content for item i

F_i - Fat content for item i

minimise $\sum_{i=1}^{10} c_i x_i$

subject to:

$$\sum_{i=1}^{10} (Va)_i x_i \geq 100$$

$$\sum_{i=1}^{10} (Vb)_i x_i \geq 120$$

$$\sum_{i=1}^{10} (Vc)_i x_i \geq 200$$

$$\sum_{i=1}^{10} (Vd)_i x_i \geq 80$$

$$\sum_{i=1}^{10} F_i x_i \leq 300$$

$$x_i \geq 0, \forall i$$

Please see the notebook "Q6 Diet problem - parts a, b and c", Part A; you should get the following output for the answer.

Sensitivity: Internal

```
Optimal solution found.


x =

         0
    9.0909
         0
         0
    1.4286
         0
         0
         0
         0
    2.2727

                        Constraint 1 is binding
fval =                  Constraint 2 is binding
   93.2468              Constraint 3 is binding
```

b)

a_i – attractiveness of item i

Replace the objective with

$$\text{maximise} \qquad \sum_{i=1}^{10} a_i x_i$$

Problem is unbounded, we can pick items with no fat content!!

Please see the notebook "Q6 Diet problem - parts a, b and c", Part B; you should get an output telling you that the problem is unbounded.

c)

Introduce extra constraints.

$$\sum_{i=1}^{10} (Va)_i x_i \leq (1.2) * 100$$

$$\sum_{i=1}^{10} (Vb)_i x_i \leq (1.2) * 120$$

$$\sum_{i=1}^{10} (Vc)_i x_i \leq (1.2) * 200$$

$$\sum_{i=1}^{10} (Vd)_i x_i \leq (1.2) * 80$$

$$\sum_{i=1}^{10} F_i x_i \geq \frac{300}{1.2}$$

Please see the notebook "Q6 Diet problem - parts a, b and c", Part C; you should get the following output for the answer (maximum attractiveness). The notebook also minimises the cost under these limits: the cost rises from £93.25 to £136.52.

Sensitivity: Internal

```
x =

    0.1937
         0
         0
         0
         0
    7.6296
    0.8593
    5.9259
         0
    7.1613

fval =

 -149.6615
```

```
Constraint 2 is binding
Constraint 4 is binding
Constraint 5 is binding
Constraint 8 is binding
Constraint 10 is binding
```

Sensitivity: Internal